EUNOIA JUNIOR COLLEGE
JC2 Preliminary Examination 2021
General Certificate of Education Advanced Level
Higher 1

CHEMISTRY

8873/01

Paper 1 Multiple Choice

23 September 2021

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.
Write your name, civics group and registration number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this question paper.
The use of an approved scientific calculator is expected, where appropriate.

This document consists of **12** printed pages.

1 Which electronic configuration represent an atom of an element, at ground state, that forms a stable ion with a charge of +3?

A $1s^2\ 2s^2\ 2p^3$

B $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$

C $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

D $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^2\ 4s^1$

2 Which particle has more protons than electrons and more protons than neutrons?
$\left(D = {}^{2}_{1}H\right)$

A H_3O^+ **B** D_3O^+ **C** D_2O **D** OH^-

3 Gaseous particle **R** has a proton number *n* and forms a stable monoatomic ion of charge –1.

Gaseous particle **S** has a proton number of (*n*+2) and it forms a stable monoatomic ion which is isoelectronic with the ion of **R**.

Which of the following statement is correct?

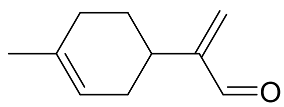
A Ion of **S** has a smaller ionic radius than ion of **R**.

B Ion of **R** releases more energy than ion of **S** when an electron is added to each particle.

C **R** has a larger atomic radius than **S**.

D **S** requires less energy than **R** when an electron is removed from each particle.

4 Limonene aldehyde is a fragrance additive in cosmetic products.



Which statements are true about the molecule?

1 Hydrogen bonds are formed between the molecules.

2 The molecule dissolves in organic solvent easily via instantaneous dipole−induced dipole interactions.

3 The bond angles about all carbon atoms in the molecule are the same.

A 1 and 2 **B** 2 and 3 **C** 2 only **D** 3 only

5 In a single covalent bond, one pair of electrons is shared by two atoms. Which statement is true for **all** single covalent bond?

A Covalent bonds are formed when bonding electrons are shared equally between two atoms.

B Covalent bonds allow the atoms involved to complete their outer electron shell configuration.

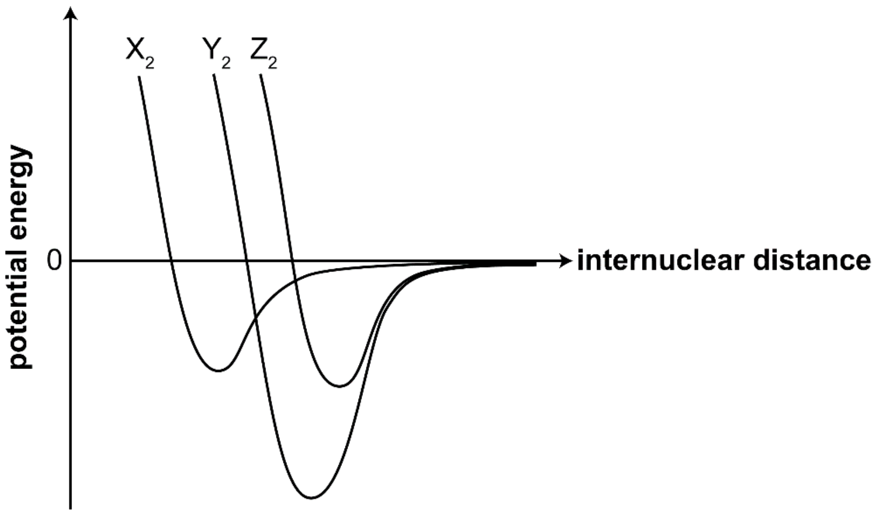
C Covalent bonds formed between atoms of different electronegativities are stronger.

D Covalent bonds formed between overlap of larger and more diffused orbitals are weaker.

6 *Use of the Data Booklet is relevant to this question.*

A Morse curve shows how the energy of a two atom system changes as a function of internuclear distance. The covalent bond length is determined by the distance at which the lowest potential energy is achieved. The lower the potential energy, the stronger the bond.

The Morse curve for three diatomic molecules, X_2, Y_2 and Z_2 are shown below.



Which of the following correctly identifies the diatomic molecules, X_2, Y_2 and Z_2?

	X_2	Y_2	Z_2
A	H_2	N_2	O_2
B	H_2	O_2	N_2
C	N_2	O_2	H_2
D	O_2	H_2	N_2

7 The melting point of lithium is 190 ºC while that of magnesium is 639 ºC. Which statement is more relevant in explaining this difference?

A The magnesium atom is larger than the lithium atom.

B The magnesium atom is less electronegative than the lithium atom.

C The magnesium ion has higher charge than the lithium ion.

D The magnesium ion contains more electrons than the lithium ion.

8 Both carbon and silicon forms tetrachloride, CCl_4 and $SiCl_4$ respectively.

The tetrachlorides were added to water separately. Only the solution from $SiCl_4$ produced an acidic solution.

Which statements are true about the reaction of the above tetrachlorides with water?

1 SiO_2 formed from hydrolysis made the solution acidic.

2 $SiCl_4$ undergoes hydrolysis due to the vacant and energetically accessible d orbitals in Si.

3 CCl_4 is inert to hydrolysis as C−Cl bond is stronger that of Si−Cl.

A 1, 2 and 3	**B** 1 and 3	**C** 2 and 3	**D** 2 only

9 Two test tubes, each containing a sample of hydrogen chloride and hydrogen bromide, were heated. A colour change was observed for only one of the test tubes.

What best explain for the observation?

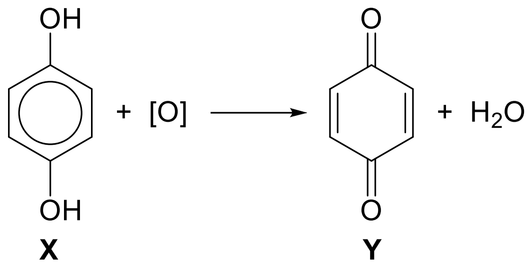
A HBr is more volatile than HCl.

B Br_2(g) is more volatile than Cl_2(g).

C HBr is a stronger reducing agent than HCl.

D H−Br bond is weaker than H−Cl bond.

10 A 1.0 g sample of **X** (M_r = 110.0) is oxidised to form **Y**.



[O] represents the oxidising agent used in the reaction.

What is the mass of **Y** formed from the above reaction?

A 0.964 g **B** 0.981 g **C** 1.00 g **D** 1.04 g

11 In the analysis of blood sample, an anticoagulant was added to form a complex ion, $[Ca(C_5H_6NO_4)_n]^{2-}$. The percentage by mass of the complex is:

Ca, 12.2%; C, 36.6%; H, 3.7%; N, 8.5%; O, 39.0%

What is the value of *n*?

A 1 **B** 2 **C** 3 **D** 4

12 Sulfur dioxide gas decolourises acidified potassium manganate(VII) solution when bubbled through the solution, forming a sulfur-containing product.

In an experiment, 20.25 cm^3 of 0.0100 mol dm^{-3} potassium manganate(VII) was required to react with 12.15 cm^3 of sulfur dioxide gas at room temperature.

$$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$$

What is the identity of the sulfur-containing product?

A S^{2-} **B** SO_3^{2-} **C** $S_2O_6^{2-}$ **D** SO_4^{2-}

13 Which of the following reactions represents the standard enthalpy change of formation of aluminium sulfate, $Al_2(SO_4)_3$?

A $2Al(s) + 3S(s) + 6O_2(g) \rightarrow Al_2(SO_4)_3(s)$

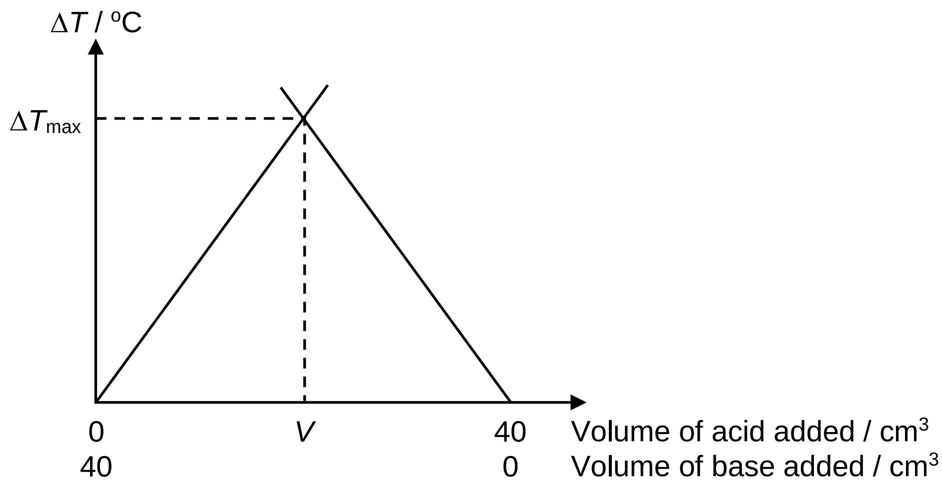
B $2Al(g) + 3S(g) + 12O(g) \rightarrow Al_2(SO_4)_3(s)$

C $2Al^{3+}(g) + 3SO_4^{2-}(g) \rightarrow Al_2(SO_4)_3(s)$

D $2Al^{3+}(g) + 3SO_4^{2-}(g) \rightarrow Al_2(SO_4)_3(g)$

14 Different volumes of HC*l*(aq) and NaOH(aq) were mixed and the change in temperature, ΔT, was measured and the results are plotted in the graph below. ΔT_{max} is the maximum temperature change measured.

Total volume of each reaction mixture was kept constant.
Both acid and base used are of the same concentration.



Which statement is not true about the experiment?

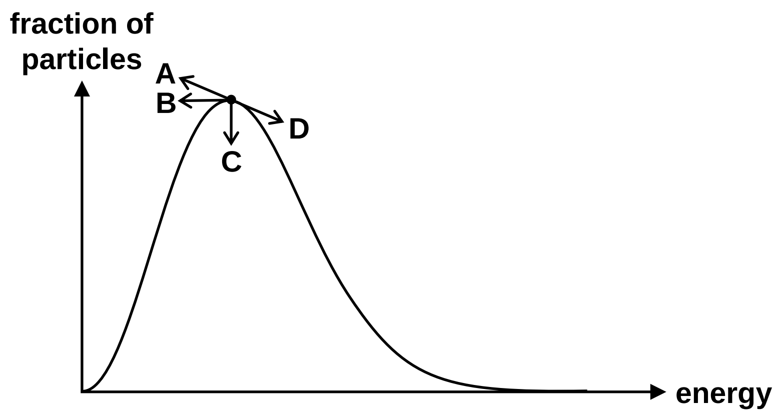
A Before the ΔT_{max}, the acid is the limiting reagent.

B V represents the point when stoichiometric amounts of acid and base react completely with each other.

C When the same concentration of NH_3(aq) is used instead, ΔT_{max} will decrease while V remains the same.

D When the same concentration of H_2SO_4(aq) is used instead, V remains the same while ΔT_{max} increases.

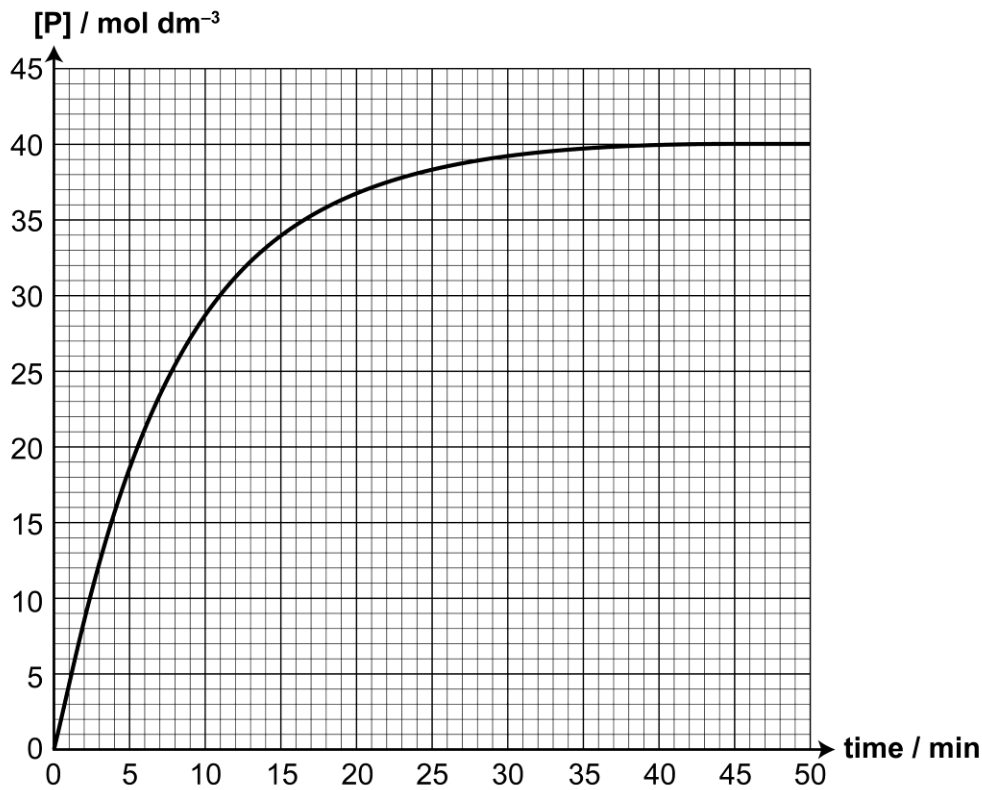
15 The Maxwell-Boltzman distribution curve of a reaction at T K is shown below. How will the peak of the distribution curve change when temperature is lowered to (T – 10) K?



16 Product **P** can be formed by reacting compound **R** under suitable conditions.

$$\mathbf{R} \rightarrow \mathbf{P}$$

In a kinetic study, the formation of product **P** was monitored as shown below.



What conclusions can be drawn from the graph?

1 The half-life of the reaction is about 2.5 min.

2 The rate constant, k, is 0.126 min^{-1}.

3 The reaction is first order with respect to **P**.

A 1 only **B** 2 only **C** 1 and 3 **D** 2 and 3

17 Catalyst is often added to speed up the rate of reaction.

Which of the following is not true of a catalyst?

A The activation energy of the reaction is lowered in the presence of a catalyst.

B When a catalyst is added, the rate constant of the reaction is increased.

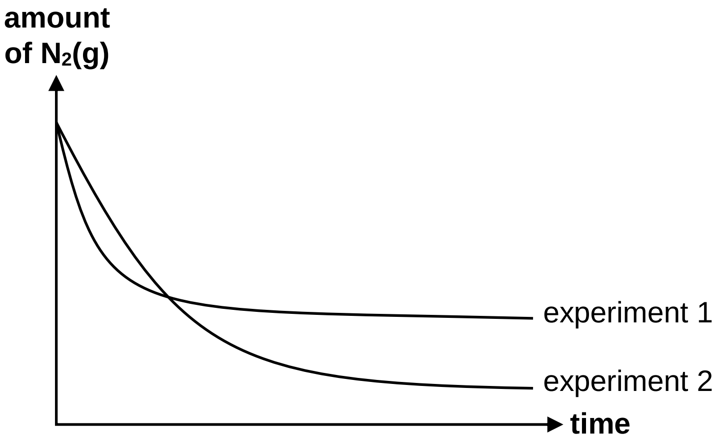
C A catalyst remains chemically unchanged throughout the reaction.

D A catalyst is added in small amounts as it is regenerated at the end of the reaction.

18 Ammonia gas is produced by reacting nitrogen gas and hydrogen gas and has the equilibrium shown:

$$N_2(g) + H_2(g) \rightleftharpoons NH_3(g) \quad \Delta H < 0$$

The change in amount of nitrogen was monitored across time for two experiments. Experiment 1 is conducted at 450 ºC and 250 atm while experiment 2 was carried out under a different set of conditions.



What change in conditions would result in experiment 2?

A Finely divided iron catalyst was added.

B Larger amount of nitrogen gas was reacted.

C Pressure of the reaction was lowered to 100 atm.

D Temperature of the reaction was lowered to 350 ºC.

19 Which of the following statements about a reaction at dynamic equilibrium are true?

1 The rate of forward reaction and rate of backward reaction is zero.

2 Equilibrium can only be achieved in a closed system.

3 Equilibrium can be established from either direction.

A 1 only **B** 1 and 2 **C** 2 and 3 **D** 1, 2 and 3

20 Which pair contains an Arrhenius base and a Brønsted acid?

A NH_3 and NH_4^+

B $Ba(OH)_2$ and H_3O^+

C KOH and CH_3COO^-

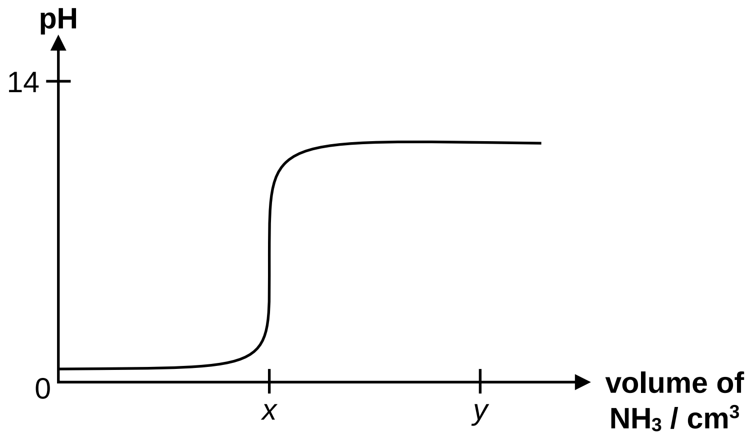
D Na_2CO_3 and HCl

21 When a strong base is titrated against a weak acid, a drastic increase in pH from pH 8 to 10 is observed

Which indicator would not be suitable for this titration?

	indicator	working pH range
A	litmus	6.0 – 8.0
B	cresol Purple	7.4 – 9.0
C	phenolphthalein	8.2 – 10.0
D	thymolphthalein	9.3 – 10.5

22 A 25 cm^3 of 0.1 mol dm^{-3} aqueous sulfuric acid was titrated with aqueous ammonia of the same concentration and the following graph is obtained.
(The graph is not drawn to scale.)



Which of the following statements is true?

A The initial pH of the solution is pH 1.

B Equivalence point for this titration is at pH 7.

C When y cm^3 of NH_3 is added, a buffer solution is obtained.

D The volume of NH_3 at x is 25 cm^3.

23 What is the total number of constitutional and cis-trans isomers a non-cyclic compound, C_5H_{10}, can have?

A 4 **B** 5 **C** 6 **D** 7

24 A straight chain organic molecule, $CH_2CHCH(OH)CN$, contains a nitrile, $-C\equiv N$ functional group.

Which of the following regarding the molecule is correct?

1 It has 10 σ bonds and 3 π bonds.

2 It has an alkene functional group.

3 It can undergo reaction with chlorine gas in the presence of uv light.

A 2 only **B** 1 and 2 **C** 2 and 3 **D** 1, 2 and 3

25 *Use of the Data Booklet is relevant to this question.*

1-bromopropane is reacted separately with the following:

reagent and condition	**organic product**
NaOH(aq), heat	X
NaOH in ethanol, heat	Y
limited Cl_2 (g), uv light	Z

Rank the products, in terms of relative molecular masses, in ascending order:

	Smallest $\longrightarrow$		**Largest**
A	X	Y	Z
B	Y	Z	X
C	Y	X	Z
D	Z	Y	X

26 In an attempt to produce butanoic acid, a student added excess butan-2-ol to a heated solution of potassium dichromate(VI) and sulfuric acid.

The resultant mixture did not contain any butanoic acid. Why is it so?

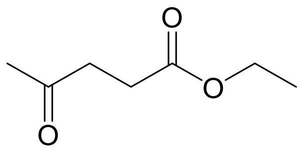
A Butan-2-ol is resistant towards oxidation.

B Butan-2-ol forms butan-2-one which cannot be further oxidised.

C Potassium dichromate(VI) is not strong enough to fully oxidise alcohol.

D Excess butan-2-ol further react with butanoic acid, resulting in a corresponding ester instead.

27 Compound **G**, $C_7H_{12}O_3$, is a fuel additive that reduces soot formation in diesel engines.

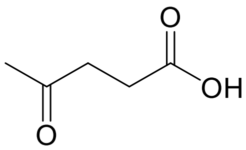
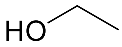


compound **G**

What is correct about compound **G**?

A 19.3 dm^3 of oxygen gas is required for complete combustion of 0.1 mol of **G** at s.t.p.

B 1 mole of **G** reacts 2 moles of hydrogen gas in the presence of nickel catalyst.

C When **G** is heated with acidified potassium manganate(VII), the only organic products formed are  and 

D A constitutional isomer of **G** has only one alcohol functional group and one carboxylic acid group.

28 Which of the following is a characteristic of thermosetting polymers?

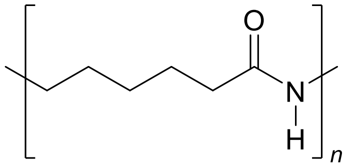
A High tensile strength and high flexibility.

B Heavily branched cross linked polymers.

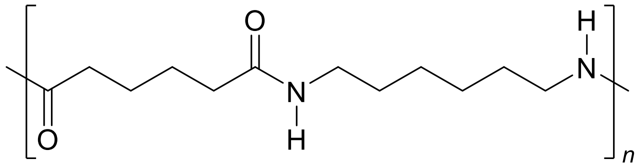
C Linear slightly branched long chain molecules.

D Soften on heating and harden on cooling, can be reused.

29 Both Nylon 6 and Nylon 66 are polyamides.



nylon 6



nylon 66

Which of the following is true?

A They can be used to make wrinkle free fabric as both polymers are not able to form hydrogen bond between its polymer chains.

B Both polymers are non-biodegradable as there are non-polar C−C bonds present.

C The formation of nylon 66 results in the elimination of water molecules.

D Both polymers are formed from identical monomers.

30 In which contexts is surface area not applicable?

A High tensile strength of graphene

B The ability of a gecko to climb a wall

C Catalytic convertor in automobiles

D Silver nanoparticles in water filtration system